

Determinacy and Confluence for $\rightarrow_c$

The execution relation $\rightarrow_c$ for While is **deterministic**: that is, for all

C, s, C_1, s_1, C_2, s_2 ,

if $\langle C, s \rangle \rightarrow_c \langle C_1, s_1 \rangle$ and $\langle C, s \rangle \rightarrow_c \langle C_2, s_2 \rangle$ then

$\langle C_1, s_1 \rangle = \langle C_2, s_2 \rangle$.

Thus, the relation $\rightarrow_c$ is **confluent**: that is, for all C, s, C_1, s_1, C_2 ,

if $\langle C, s \rangle \rightarrow_c^* \langle C_1, s_1 \rangle$ and $\langle C, s \rangle \rightarrow_c^* \langle C_2, s_2 \rangle$ then there exists

$\langle C', s' \rangle$ such that $\langle C_1, s_1 \rangle \rightarrow_c^* \langle C', s' \rangle$ and $\langle C_2, s_2 \rangle \rightarrow_c^* \langle C', s' \rangle$

Analogous results hold for the relations $\rightarrow_e$ and $\rightarrow_b$.